5	**(a)(i)**	out of the page	B1
	(a)(ii)	magnetic force provides centripetal force: $Bev = mv^2/r$ $v = Ber / m$	B1 B1
	(b)(i)	gain in kinetic energy = ½ $m\ v^2 = eV$ <u>OR</u> work done = eEd = ½ $m\ v^2$	B1
	(b)(ii)	[use $v = Ber / m$ and ½ $m\ v^2 = eV$ to obtain] $r^2 = 2Vm / eB^2$ OR $r = [2Vm / eB^2]^{1/2}$	 C1
		correct substitution for V, m, e and B^2	C1
		$r = 1.6 \times 10^{-2}$ m	A1
	(c)	the force acting on the electron is always perpendicular to its direction of motion no work is done on the electron (thus no change in its kinetic energy or speed)	B1 B1